

# GENERATIVE AGENTS IN AGENT-BASED MODELING: OVERVIEW, VALIDATION, AND EMERGING CHALLENGES

CSQ413 Seminar Presentation

**Submitted by:**

ARAVIND A KAMATH (CEC23CS041)

**Under the Guidance of:**

Mrs. JAYASREE K

Assistant Professor

Department of Computer Science and Engineering  
College of Engineering, Cherthala

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# Introduction: Agent-Based Modeling (ABM)

- **Bottom-Up Paradigm:** Simulates autonomous entities (agents) interacting locally within a shared environment.
- **Emergence Principle:** Macro-level patterns emerge naturally from decentralized individual decisions.
- **Wide Applicability:** Used in traffic planning, epidemic spreading and financial market dynamics.
- **Core Objective:** Explains complex collective phenomena without requiring central control.

# Limitations of Classical Behavioral Models

- **Strict Economic Rationality:** Assumes agents are perfectly rational utility maximizers (*homo economicus*).
- **Absence of Emotional Nuance:** Cannot capture human stress, fear, empathy or shifting moods.
- **No Natural Language:** Communication is restricted to numbers and binary flags.
- **Rigid Decision Rules:** Fixed if-then heuristics fail when unprecedented situations arise.
- **Population Homogeneity:** Agents are grouped into simple types, ignoring individual diversity.

# The Concept of 'Generative' in Social Science

- **Epstein's Principle (1999):** *"If you didn't grow it, you didn't explain it."*
- **Classical Meaning:** "Generative" meant growing macro-structures from simple local rules without central control.
- **Foundational Models:** Schelling's segregation model (1971) and Reynolds' flocking boids (1987).
- **The Technological Shift:** Classical models used hand-coded rules; modern GABMs use Large Language Models for cognitive reasoning.

# Traditional ABM vs. Generative ABM: Urban Traffic Flow

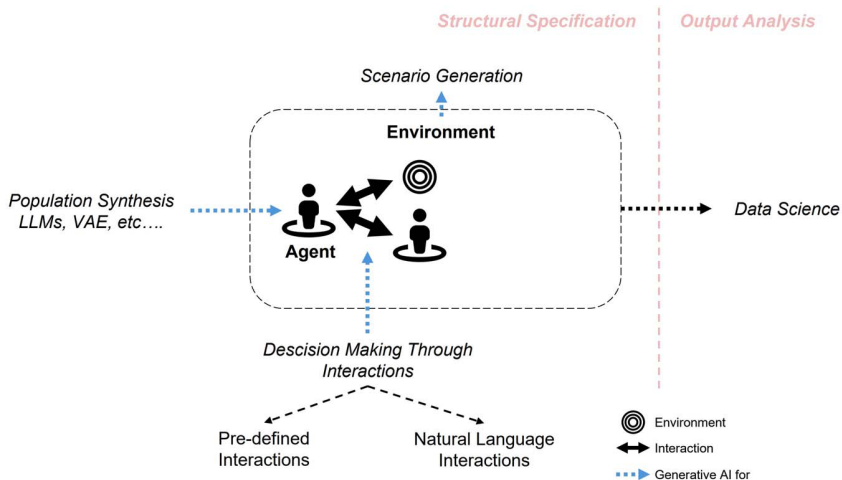
**Table:** Comparison between Traditional ABM and Generative ABM (Table I)

| Dimension      | Traditional ABM                            | Generative ABM (GABM)                           |
|----------------|--------------------------------------------|-------------------------------------------------|
| Scenario       | Standard vehicles following signals.       | Drivers influenced by stress and habits.        |
| Objective      | Maximize throughput and find bottlenecks.  | Study how emotions and norms shape traffic.     |
| Outcomes       | Local signal timings; hard to generalize.  | Generalizable behavioral insights.              |
| Decision Basis | Fixed shortest-path heuristics (Dijkstra). | Contextual reasoning (safety, stress, comfort). |

# Precursor Agent Architectures: EAs and RL

- **Evolutionary Algorithms (EAs):** Optimize agent rules via selection, crossover and mutation across generations.
- **EA Bottlenecks:** High computational cost, premature local convergence and heavy parameter tuning.
- **Reinforcement Learning (RL):** Optimizes numerical policies to maximize reward signals through trial and error.
- **RL Bottlenecks:** Sample inefficiency, multi-agent instability and total absence of language dialogue.
- **Takeaway:** Neither EAs nor RL provide human-like commonsense reasoning or open-ended communication.

# AI Integration in ABM Development



**Figure:** AI integration across Structural Specification and Output Analysis (Source: Base Paper Figure 1).

# Roles of Generative AI in ABM Development

- **Population Synthesis:** Generates diverse agent profiles with realistic ages, traits and occupations.
- **Interactive Decision Making:** LLMs act as cognitive engines, deciding actions dynamically via dialogue.
- **Scenario Generation:** Synthesizes dynamic environments, policy interventions and social events.
- **Macro-Level Tracking:** Uses machine learning to detect patterns and emergent trends across runs.
- **Direct Agent Interviewing:** Allows observers to query agents directly in natural language about their choices.

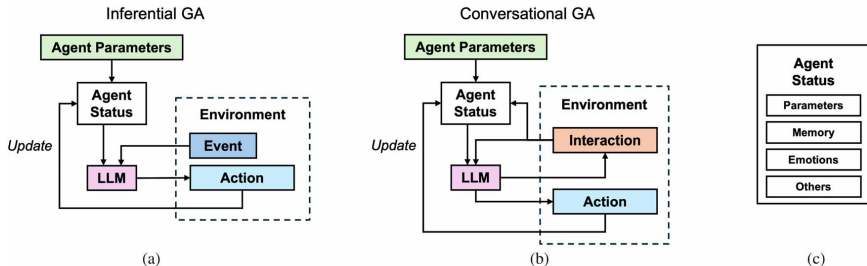
# Defining Generative Agents (GAs)

- **Formal Definition:** Autonomous computational entities powered by Large Language Models capable of human-like behavior and dialogue.
- **Mechanical Layer:** Rule-based physical substrate handling spatial coordinates, collisions and clock time.
- **Cognitive Layer:** LLM processes sensory perceptions, queries memory streams and decides actions.
- **Emergent Behaviors:** Decisions arise from pre-trained world knowledge rather than hand-coded if-then rules.

# Internal State: Static Parameters vs. Dynamic Status

- **Static Parameters (Identity Anchor):** Demographic data (age, gender, job), Big Five OCEAN personality traits and social network ties.
- **Dynamic Status (Runtime Representation):** Physical coordinates, current tasks, active goals and emotional affect.
- **Decoupled Architecture:** Separates internal state database storage from the LLM prompt engine.
- **Affective Tracking:** Records sentiment shifts, relationship changes and recent interaction history.

# GA Decision Architectures: Feedback Loop and Status



**Figure:** LLM-based feedback loop architectures for Generative Agents (Source: Base Paper Figure 2).

# Inferential vs. Conversational Feedback Loops

- **Inferential Loop [Figure 2(a)]:** Agent receives an environmental event, checks internal status, and the LLM outputs a discrete physical action.
- **Environment Update:** Discrete actions directly alter physical variables in the simulation world.
- **Conversational Loop [Figure 2(b)]:** Agent receives dialogue from peers, updates internal beliefs and status, and outputs natural language text.
- **Social Communication:** Natural language responses directly influence the mental states of neighboring agents.

# Core Cognitive Modules: Memory Stream & Retrieval

- **Memory Stream:** Chronological database logging perceptions, dialogues and thoughts in natural language text.

$$\text{Score}(m, q) = \alpha \cdot \text{Recency}(m) + \beta \cdot \text{Importance}(m) + \gamma \cdot \text{Relevance}(m, q)$$

## Symbol Definitions:

- $\text{Score}(m, q)$ : Total retrieval priority for memory record  $m$  under situational query  $q$ .
- $\alpha, \beta, \gamma$ : Non-negative weights balancing recency, importance and relevance.
- $\text{Recency}(m) = e^{-\lambda \Delta t}$ : Exponential decay with decay rate  $\lambda$  and elapsed time  $\Delta t$ .
- $\text{Importance}(m)$ : Integer score (1 to 10) assigned by the LLM for event significance.
- $\text{Relevance}(m, q) = \frac{\mathbf{e}_m \cdot \mathbf{e}_q}{\|\mathbf{e}_m\| \|\mathbf{e}_q\|}$ : Cosine similarity between embedding vectors  $\mathbf{e}_m$  and  $\mathbf{e}_q$ .

# Core Cognitive Modules: Reflection & Dynamic Planning

- **Reflection Hierarchy:** Periodically synthesizes low-level memory records into higher-order abstract thoughts.
- **Belief Tree:** Forms an evolving directed tree from raw observations (leaves) to generalized beliefs (roots).
- **Recursive Planning:** Decomposes broad daily agendas into hourly schedules, then minute-level action sequences.
- **Dynamic Reactive Re-planning:** Pauses routine execution and updates plans when unexpected events occur.

# Literature Survey: Existing GABMs (Part 1)

**Table:** Literature survey of foundational Generative Agent platforms (Source: Base Paper Table II)

| Paper & Author                                 | Scenario                                 | Core Architecture                                     | Key Highlights                                      | Validation Approach                                     |
|------------------------------------------------|------------------------------------------|-------------------------------------------------------|-----------------------------------------------------|---------------------------------------------------------|
| Generative Agents<br><i>Park et al. (2023)</i> | 25 agents in 2D RPG Smallville town.     | Memory stream + Reflection tree + Daily planning.     | Emergent party planning and information spread.     | 100 human judges across 5 interview categories.         |
| Concordia<br><i>Vezhnevets et al. (2023)</i>   | Grounded physical and social spaces.     | Modular language agents with centralized Game Master. | Enforces physical reality and resource constraints. | Qualitative analysis and design guidelines.             |
| SOTOPIA<br><i>Zhou et al. (2024)</i>           | Goal-driven cooperation and negotiation. | Interactive environment with distinct agent profiles. | Quantitative social intelligence benchmarking.      | Dual evaluation: Humans vs. LLM-as-a-judge (7 metrics). |

# Literature Survey: Existing GABMs (Part 2)

**Table:** Literature survey of social simulation and domain-specific GABMs (Source: Base Paper Table II)

| Paper & Author                                | Scenario                                   | Core Architecture                                | Key Highlights                                  | Validation Approach                           |
|-----------------------------------------------|--------------------------------------------|--------------------------------------------------|-------------------------------------------------|-----------------------------------------------|
| CRSEC Norms<br><i>CRSEC (2024)</i>            | Norm emergence with 10 GAs in Smallville.  | 4 stages: Creation, Spread, Evaluate, Comply.    | Self-governing rules (indoor smoking ban).      | 30 human evaluators testing compliance rates. |
| S3 Platform<br><i>Gao et al. (2023)</i>       | Digital social media and opinion dynamics. | LLM agents with attitude and emotion tracking.   | Replicates retweet cascades and echo chambers.  | Real-world social network dataset matching.   |
| Epidemic GAs<br><i>Williams et al. (2023)</i> | Disease outbreak and emergency decisions.  | Inferential GAs driven by Big Five OCEAN traits. | Personality influence on quarantine compliance. | Empirical matching against epidemic curves.   |

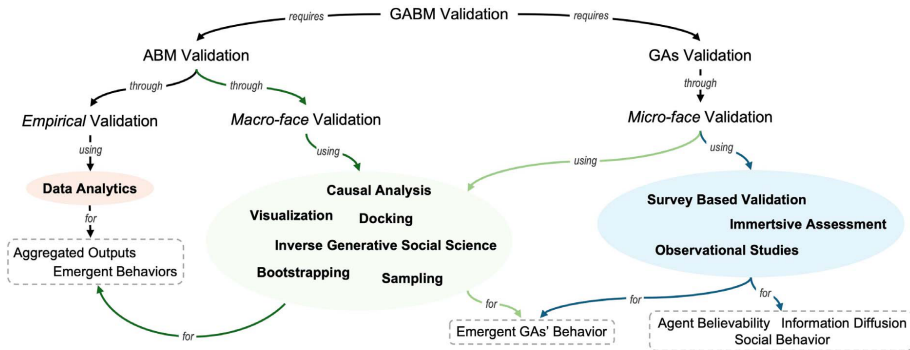
# The Fundamental Validation Dilemma

- **Purpose of Validation:** Ensures simulation models reflect real-world social processes rather than software bugs.
- **White-Box Traditional ABMs:** Model rules and transitions can be inspected and verified mathematically.
- **Black-Box Generative Agents:** Foundation models produce non-deterministic outputs from identical initial prompts.
- **Evidence Hierarchy:** Adapts evidence standards: controlled trials (highest), observational data (middle) and face validation (lowest).

# Foundational and Face Validation in Traditional ABM

- **Data Analytics & Docking:** Verifies distributions, correlations and compares outputs against independent models.
- **Empirical Testing & Resampling:** Validates predictions on historical data and uses bootstrapping for confidence bounds.
- **Microface Validation:** Checks individual agent decisions and emotional logic for plausibility.
- **Macroface Validation:** Confirms collective group patterns (crowd flows, segregation) match real-world intuition.
- **Immersive Probing:** Domain experts enter virtual dashboards or VR to observe and interview agents.

# GABM Validation Workflows Architecture



**Figure:** Validation workflows for GABMs across different validation purposes (Source: Base Paper Figure 3).

# Integrated Validation Pathways

- **Empirical Pathway (Quantitative Matching):** Conditions initial demographics with census data and tests aggregate trajectories against historical data.
- **Model Alignment:** Applies data analytics and docking across runs to ensure numerical reliability.
- **Macroface Pathway (Collective Patterns):** Validates emergent group dynamics such as spatial clustering and norm spread without central control.
- **Microface Pathway (Individual Realism):** Probes individual agents through structured interviews to check reasoning, memories and consistency.

# Automated Validation: LLM-as-a-Judge & Algorithmic Fidelity

- **LLM-as-a-Judge:** Uses independent LLMs to interview agents and grade multi-agent dialogues at large scale.
- **Evaluation Strengths:** High accuracy on objective goals, factual consistency and quantitative criteria.
- **Evaluation Limits:** Struggles with subtle social nuances, passive aggression, sarcasm and deception.
- **Algorithmic Fidelity:** Measures how accurately simulated personas reflect genuine human demographic attitudes.
- **Similarity Principle:** Rigorous testing on sample scenarios builds confidence across related unobserved spaces.

# Comparison: Resources and Agent Design

**Table:** Comparison of ABM Paradigms: Resources and Agent Design (Table III)

| Evaluation Dimension             | Rule-Based      | EA-Based     | RL-Based   | GABM         |
|----------------------------------|-----------------|--------------|------------|--------------|
| <b>Resources</b>                 |                 |              |            |              |
| Computational Efficiency         | High            | Moderate     | Low        | Low          |
| Scalability (Millions of Agents) | High            | Moderate     | Low        | Very Low     |
| Data or Domain Knowledge         | Dependent       | Dependent    | Dependent  | Data-Less    |
| <b>Agent Design</b>              |                 |              |            |              |
| Control over Agents              | Strict / Direct | High         | Moderate   | Low          |
| Learning During Simulation       | Static          | Evolutionary | Continuous | Impractical  |
| Real-Time Optimization           | Not Needed      | Required     | Required   | Not Needed   |
| Hyperparameter Tuning            | Minimal         | Heavy        | Heavy      | Prompt-Based |
| Fast Prototyping                 | Labor-Intensive | Moderate     | Slow       | Very Rapid   |

- **Resource Trade-Off:** Rule-based models scale to millions; GABMs require heavy computation.
- **Design Trade-Off:** GABMs enable rapid prototyping with prompts, but offer less direct control.

# Comparison: Agent Behavior and Validation

**Table:** Comparison of ABM Paradigms: Behavior and Validation (Table III)

| Evaluation Dimension         | Rule-Based    | EA-Based        | RL-Based          | GABM               |
|------------------------------|---------------|-----------------|-------------------|--------------------|
| <b>Agent Behavior</b>        |               |                 |                   |                    |
| Determinism                  | Deterministic | Stochastic      | Stochastic        | Stochastic         |
| Reliable Decisions           | Guaranteed    | Suboptimal Risk | Early Instability | Hallucination Risk |
| Cognitive Functions          | None          | None            | Primitive         | Advanced           |
| Natural Language Interaction | None          | None            | None              | Human-like         |
| Nuanced Mental Models        | None          | None            | None              | Beliefs / Emotions |
| <b>Validation</b>            |               |                 |                   |                    |
| Decision Transparency        | White-Box     | Traceable       | Black-Box         | Black-Box          |
| Validation Efficiency        | High          | Moderate        | Quantitative      | Resource-Intensive |
| Direct Human Interaction     | No            | No              | No                | Direct Interview   |

- **Behavioral Depth:** GABMs uniquely support rich cognitive models and natural language dialogue.
- **Validation Challenge:** Traditional models offer transparent rules; GABMs are opaque black boxes.

# Challenge 1: Large Language Model Hallucinations

- **Factuality Hallucinations:** Generating assertions that contradict verified physical environment facts.
- **Faithfulness Hallucinations:** Diverging from assigned persona values, moral codes or prior conversation history.
- **Data Stage Mitigation:** Grounding prompts with Retrieval-Augmented Generation (RAG) using live environment facts.
- **Training Stage Mitigation:** Applying Reinforcement Learning from Human Feedback (RLHF) and persona fine-tuning.
- **Inference Stage Mitigation:** Enforcing constrained grammar decoding (e.g., JSON schema) and multi-agent peer verification.

## Challenge 2: Computational Bottlenecks and Scaling

- **Computational Bottlenecks:** Quadratic attention latency and pairwise agent communication scaling:  
 $\text{Complexity}_{\text{Attention}} = \mathcal{O}(n^2 \cdot d)$ ,  $\text{Complexity}_{\text{Communication}} = \mathcal{O}(N^2)$

### Symbol Definitions:

- $n$ : Sequence length (number of input context tokens in memory prompt).
- $d$ : Embedding representation dimension of the Transformer model.
- $N$ : Total number of interacting generative agents in the simulation.
- $M$ : Number of memory records searched during vector retrieval ( $\mathcal{O}(\log M)$  to  $\mathcal{O}(M)$ ).
- **The Scalability Gap:** Traditional ABMs run  $10^6$  agents; pure GABMs remain limited to dozens or hundreds.

# Discussion: Cities as a Natural Testbed for GABMs

- **Alignment with Real-World Text:** Urban laws, historical archives and census data closely match LLM training text.
- **Beyond Physical Metrics:** Traditional urban models measure only functional data (traffic delay, throughput, capacity).
- **Qualitative Experience Mapping (QEM):** Models subjective human feelings: safety perceptions in unlit streets, comfort and transit stress.
- **Human-Centered Planning:** Informs urban policy to support community emotional well-being and pedestrian safety.

# The Proposed Solution: Combined Hybrid ABM-GABM

- **Complementary Synergy:** Unifies the high computational speed of rule-based ABMs with the cognitive depth of GABMs.
- **Low-Level Mechanistic Layer (Traditional ABM):** Handles deterministic, high-frequency physics: collision avoidance and kinematics ( $\mathcal{O}(1)$  or  $\mathcal{O}(N)$ ).
- **High-Level Cognitive Layer (Generative Agents):** Invoked only during critical decision points: moral choices, negotiations and social cooperation.
- **Key Benefit:** Resolves computational bottlenecks while preserving rich human-like behavior.

# Case Study: Building Evacuation & Hybrid Validation

- **Emergency Building Fire Evacuation:** Illustrates the division of labor in an emergency crisis.
- **Low-Level Rule Engine:** Moves agents through corridors, stairwells and exit doors using fast physics heuristics.
- **High-Level LLM Engine:** Decides whether an agent stops to help an injured peer or flees to save itself.
- **Macro-Validation:** Sensor data confirms crowd evacuation clearance times match building drill records.
- **Micro-Validation:** Post-simulation interviews verify individual survival reasoning and moral choices.

# Summary of Key Insights and Contributions

- **Cognitive Paradigm Shift:** Replaced rigid rational heuristics with believable, conversational agents.
- **Conceptual Clarity:** Distinguished Epstein's classic generative social science from modern generative AI.
- **Integrated Validation Blueprint:** Formulated a multi-tiered pipeline combining empirical data, face validation and LLM judges.
- **Hybrid Path Forward:** Coupled traditional physics with cognitive LLMs to achieve scalability and psychological realism.

# Future Research Trajectories

- **Small Language Models (SLMs):** Fine-tuning compact models to reduce inference latency and computational costs.
- **Standardized Benchmarks:** Establishing open-source multi-agent test environments for objective comparison.
- **Hallucination Bounds:** Formulating mathematical bounds on hallucination rates for policy simulations.
- **Ethical Safeguards:** Preventing bias propagation and harmful stereotypes in simulated human societies.

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# Thank You!

## Questions and Discussion